Numbering systems

Decimal numbers

We use the decimal system to represent numbers:

- The system uses 10 digits (hence “decimal”), from 0 to 9
- 10 is called the *base*
- Numbers are represented by using *powers of the base*
- For example, the number **1235** is:

$$
\begin{array}{rrrr}
1000\ + & 200\ + & 30\ + & 5\ = \\
\mathbf{1} \times 1000\ + & \mathbf{2} \times 100\ + & \mathbf{3} \times 10\ + & \mathbf{5} \times 1\ = \\
\mathbf{1} \times 10^3 + & \mathbf{2} \times 10^2 + & \mathbf{3} \times 10^1 + & \mathbf{5} \times 10^0
\end{array}
$$

Binary numbers

Computers store data in bits, which can take only two values: on/off, true/false and, for numbers, 0/1

- Instead of 10 digits to represent numbers, computers have two: 0 and 1
- We can represent numbers using *base 2*
  - Like decimal, with powers of 2
- Next, an example: convert the binary number **1010011** to decimal.

Binary numbers

The binary number **1010011** is, in decimal:

1 0 1 0 0 1 1

1×2^0

Binary numbers

The binary number **1010011** is, in decimal:

```
1   0   1   0   0   1   1
```

$1 \times 2^1 + \quad 1 \times 2^0$

Binary numbers

The binary number **1010011** is, in decimal:

1 0 1 0 0 1 1

$0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

Binary numbers

The binary number **1010011** is, in decimal:

1 0 1 0 0 1 1

$$0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

Binary numbers

The binary number **1010011** is, in decimal:

1 0 1 0 0 1 1

$$1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

Binary numbers

The binary number **1010011** is, in decimal:

$$1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1$$

$$0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

Binary numbers

The binary number **1010011** is, in decimal:

1 0 1 0 0 1 1

$$1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

Binary numbers

The binary number **1010011** is, in decimal:

1 0 1 0 0 1 1

$1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$

2^6

64

Binary numbers

The binary number **1010011** is, in decimal:

1 0 1 0 0 1 1

$$1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$2^6 + 2^4$$

$$64 + 16$$

Binary numbers

The binary number **1010011** is, in decimal:

$$1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1$$

$$1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

$$2^6 + 2^4 + 2^1$$

$$64 + 16 + 2$$

Binary numbers

The binary number **1010011** is, in decimal:

$$1 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1$$

$$1 \text{ x } 2^6 + 0 \text{ x } 2^5 + 1 \text{ x } 2^4 + 0 \text{ x } 2^3 + 0 \text{ x } 2^2 + 1 \text{ x } 2^1 + 1 \text{ x } 2^0$$

$$2^6 + 2^4 + 2^1 + 2^0 =$$

$$64 + 16 + 2 + 1 = \mathbf{83}$$

The problem with binary

- Binary numbers have a problem: they can be too long

- Take for example, the number **1235**. In binary, it is:

  10011010011 (You can verify this by calculating $2^{10} + 2^7 + 2^6 + \ldots$)

- It takes 11 binary digits to represent 4 decimals!

- The solution? The hexadecimal (base 16) numbering system, presented next

Hexadecimal numbers

- As we've seen, we can use any base to represent numbers
- Base 16 (hexadecimal, abbreviated "hex") is very useful
  - We'll see why soon
- To represent numbers in base 16 we need… 16 digits. We use:
  - 0 to 9, same as decimal
  - A to F, to represent 10 to 15
- For example, the hexadecimal number **24A** is, in decimal:

$$\mathbf{2} \times 16^2 + \mathbf{4} \times 16^1 + \mathbf{10} \times 16^0 = 512 + 64 + 10 = \mathbf{586}$$

Hexadecimal numbers

- Hexadecimal numbers are useful for representing binary data

- We can represent groups of 4 bits with a single hexadecimal digit:

Decimal	0	1	2	3	4	5	6	7
Binary	0000	0001	0010	0011	0100	0101	0110	0111
Hexadecimal	0	1	2	3	4	5	6	7

Decimal	8	9	10	11	12	13	14	15
Binary	1000	1001	1010	1011	1100	1101	1110	1111
Hexadecimal	8	9	A	B	C	D	E	F

Next, an example: represent decimal **1235** in hexadecimal

Hexadecimal numbers - example

- Recall that decimal **1235** is, in binary, **10011010011**
- To represent it in hexadecimal:
  - We first arrange it in groups of 4 bits from the right: **100 1101 0011**
  - then convert each 4-bit group to its corresponding hex digit:

Binary	**0100**	**1101**	**0011**
Hexadecimal	**4**	**D**	**3**

- Binary **10011010011** is **4D3** in hexadecimal
  - Much shorter!
- Next: how to convert a decimal number to hexadecimal, step by step